

SOP - (Disjunctive Normal form)

## Min-Term (SOP)

→ A product term containing all variable once.

→ It is in a complemented and uncomplemented form.

for example :

$ABC, \bar{A}\bar{B}\bar{C}, A\bar{B}\bar{C}$  etc.

## Max-Term (POS) (Conjunctive normal form)

→ A sum term containing all variables once (non-repeated) in complemented and un-complemented form.

for examples-

$A+B+C, \bar{A}+B+C, \bar{A}+\bar{B}+\bar{C},$

## Canonical or Standard form

A boolean function (expression) contain min-terms



or both) max-terms (but not standard form).

for example (SOP)

$$F(X, Y, Z) = XYZ + X\bar{Y}Z + XY\bar{Z}$$

(Sum of product form)

$$F(X, Y, Z) = (X+Y+Z)(X+\bar{Y}+Z)(X+Y+\bar{Z})$$

(SPOS product of sum form)

Using properties

$$XX' = 0$$

$$X + \bar{X} = 1$$

$$X \cdot \bar{X} = 0$$

$$X + \bar{X} = 1$$

Convert in min-term

$$F = X + YZ$$

$$\begin{aligned} F &= X(1) + YZ(1) \\ &= X(Y+\bar{Y})(Z+\bar{Z}) + YZ(X+\bar{X}) \\ &= X(YZ + Y\bar{Z} + \bar{Y}Z + \bar{Y}\bar{Z}) + XYZ + \bar{X}YZ \\ &= XYZ + XY\bar{Z} + X\bar{Y}Z + X\bar{Y}\bar{Z} + XYZ + \bar{X}YZ \\ &= XYZ + X\bar{Y}\bar{Z} + X\bar{Y}Z + \bar{X}YZ + X\bar{Y}\bar{Z} \end{aligned}$$

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SOP (Disjunctive-Normal-form)

- It is a summation of Product terms.

E.g.  $Y = AB + \bar{A}\bar{B}C + \bar{A}B\bar{C}$

POS (Conjunctive normal form)

- It is a summation of sum terms.

E.g.  $Y = (A+B) \cdot (A+C) \cdot (\bar{A}+\bar{B})$

Canonical form

- SSOP

- Each product term contains all the variables of the function.

E.g.  $= \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C}$

- SPOS

- Each sum term contains all the variables of the function.

E.g.  $= (\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+C)(A+B+\bar{C})$

Minterm

- Each individual term in SSOP is called as minterm.

(m, y, z) =  $\bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC$   
minterm

Maxterm

- Each individual term in SPOS is called as maxterm.

$= (A+B+\bar{C})(A+\bar{B}+C)$   
maxterm



SSOP to SPOS & SSPOS to SSOP Conversion

- i)  $F(A, B, C) = \sum m(0, 1, 4, 7)$   
Convert SSOP to SPOS.

Total no of variables = 3

Possible Term =  $2^3 = 8 = [0, 1, 2, 3, 4, 5, 6, 7]$   
 $F(A, B, C) = \prod M(2, 3, 5, 6)$

- ii) Express  $\prod M(2, 3, 5, 6)$  in SPOS form.

$$= (A + B + C)(A + \bar{B} + \bar{C})(\bar{A} + B + \bar{C})(\bar{A} + \bar{B} + C)$$

- iii) Express  $\sum m(0, 1, 4, 7)$  in SSOP form.

$$= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC$$

- iv)  $F(A, B, C) = \prod M(1, 2, 6)$   
Convert SPOS to SSOP.

Total no of variables = 3

Possible Term =  $2^3 = 8 = [0, 1, \dots, 7]$

$$F(A, B, C) = \sum m(0, 4, 5, 7)$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

- v)  $ABC + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C}$  Convert to SPOS.

$$= m_7 + m_0 + m_2$$

$$F(A, B, C) = \prod M(0, 1, 2, 4, 5)$$

Self dual:

→ If dual of function is same function then it is referred as self dual.

e.g  $F = AB + BC + AC$

Find given function F is self dual or not

$$F = (A+B) \cdot (B+C) \cdot (A+C)$$

$$= (A+B) \cdot (A+C) \quad \therefore [As \text{ for distributive rule } (A+B) \cdot (B+C) = A+B+C]$$

$$= BA + BC + AC + AC$$

$$= BA + BC + AC$$

$$= AB + BC + AC$$

$$F_d = F \quad \text{So, function } f \text{ is self dual.}$$

Note: for n number of variable,  
total self dual =  $2^{2^n - 1}$

Q# State Consensus Theorem in boolean algebra?

→ In simplification of boolean expression, the redundant term in an expression can be eliminated to form the equivalent expression.

→ The theorem used for this simplification is called consensus theorem.



## What is boolean Algebra?

- George boole introduced the boolean algebra.
- It is a set of rules used to minimize boolean expression without changing its functionality.

### Draw back:

- we have to use truth table to check functionality.

## What is binary logic?

- Binary logic consists of binary variables and logical operations.
- The variables are designated by alphabets such as A, B, C, x, y, z etc.; with each variable having two distinct values.

## What is Dual and self dual of boolean expression?

### Dual:

- To get dual of any given expression we need to replace

- \* OR with AND
- \* AND with OR
- \* 1 with 0
- \* 0 with 1

$$\text{e.g. } (A+B) \cdot (\bar{A}+C)(B+C)$$

$$\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{C}$$

## Convert in max-term

$$F = X + YZ$$

$$F = X + YZ$$

$$F = (X+Y)(X+Z)$$

$$\therefore (A+BC) = (A+B)(A+C)$$

$$F = (X+Y+0)(X+Z+0)$$

$$= (X+Y+Z\bar{Z})(X+Z+Y\bar{Y})$$

$$= (X+Y+Z)(X+Y+\bar{Z})(X+Z+Y)$$

$$= (X+Y+Z)(X+Y+\bar{Z})(X+\bar{Y}+Z)$$

## Convert in min Term and max Term.

$$F(P, Q, R) = \bar{P}\bar{R} + QR$$

for min-term:

$$= \bar{P}\bar{R} + QR$$

$$= \bar{P}\bar{R}(1) + QR(1)$$

$$= \bar{P}\bar{R}(Q+\bar{Q}) + QR(P+\bar{P})$$

$$= P\bar{Q}\bar{R} + P\bar{Q}\bar{R} + PQR + \bar{P}QR$$

$$= P\bar{Q}\bar{R} + P\bar{Q}\bar{R} + \bar{P}QR + P\bar{Q}\bar{R}$$

(Sop form)



Convert into max-term form (POS).

$$F(Q, P, R) = P\bar{R} + QR$$

$$= P\bar{R} + QR$$

$$= \bar{P}(\bar{R} + Q)$$

$$= (P\bar{R} + Q)(P\bar{R} + R)$$

$$= (Q + P)(Q + \bar{R})(P + R)(R + \bar{R})$$

$$= (Q + P)(Q + \bar{R})(P + R)(1)$$

$$= (Q + P + 0)(Q + 0 + \bar{R})(P + R + 0)$$

$$= (Q + P + \bar{R})(Q + P\bar{R})(P + R + Q)$$

$$= (Q + P + R)(Q + P + \bar{R})(Q + \bar{R} + P)$$

$$= (Q + P + R)(Q + P + \bar{R})(\bar{P} + Q + \bar{R})$$

$$= (Q + P + R)(Q + P + \bar{R})(\bar{P} + Q + \bar{R})$$

(POS form)

Convert the following boolean expressions to SOP form.

$$(a) AB + B(CD + EF)$$

$$= AB + B(CD + EF)$$

$$= AB + BCD + BEF$$

$$(b) (A+B)(B+C+D)$$

$$= AB + AC + AD + BB + BC + BD$$

$$(c) \overline{A+B+C}$$

$$= \overline{A+B} \cdot \bar{C}$$

$$= (A+B) \bar{C}$$

$$= A\bar{C} + B\bar{C}$$

Convert  $\bar{A}B\bar{C} + A + \bar{B}$   
( $B + \bar{C} + A\bar{B}$ )

To SOP form.

$$= \bar{A}B\bar{C} + (A + \bar{B})(B + \bar{C} + A\bar{B})$$

$$= \bar{A}B\bar{C} + AB + A\bar{C} + A\bar{B} + \bar{B}\bar{C} + 0$$

$$= \bar{A}B\bar{C} + AB + A\bar{C} + A\bar{B} + \bar{B}\bar{C}$$



Convert the following  
boolean into SSOP  
form.

$$\bar{A}\bar{B}C + \bar{A}\bar{B} + AB\bar{C}D$$

$$= \bar{A}\bar{B}C + \bar{A}\bar{B} + AB\bar{C}D$$

$$= \bar{A}\bar{B}C(1) + \bar{A}\bar{B}(1) + AB\bar{C}D$$

$$= \bar{A}\bar{B}C(D+\bar{D}) + \bar{A}\bar{B}(C+\bar{C})(D+\bar{D}) + AB\bar{C}D$$

$$= \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D} + AB\bar{C}D$$

$$= \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$+ \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + AB\bar{C}D$$

$$= \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C(D+\bar{D}) + \bar{A}\bar{B}\bar{C}\bar{D} +$$

$$\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + AB\bar{C}D$$

$$= \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} +$$

$$\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D}$$

$$+ AB\bar{C}D$$

$$= \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D}$$

$$+ \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D$$



Convert the following  
boolean expression into  
Standard SOP form.

$$(A + \bar{B} + C)(\bar{B} + C + \bar{D})(A + \bar{B} + \bar{C} + D)$$

$$= (A + \bar{B} + C + D\bar{D})(A\bar{A} + \bar{B} + C + \bar{D})$$

$$= (A + \bar{B} + C + D)(\bar{A} + \bar{B} + \bar{C} + D)$$
$$= (A + \bar{B} + C + D)(A + B + C + \bar{D})(A + \bar{B} + C + \bar{D})$$
$$(\bar{A} + \bar{B} + C + \bar{D})(\bar{A} + \bar{B} + \bar{C} + D)$$